

MATH INTERNAL ASSESSMENT

NOVEMBER 2024 SESSION

Analysing Battery Discharging in My Bluetooth Headphones through Curve
Fitting

Introduction

In a world increasingly reliant on technology, the use of portable electronic devices such as headphones and smartphones has been a part of our daily lives. Throughout the COVID-19 pandemic, technology became a lifeline, bridging distances and facilitating connectivity while maintaining social distancing. One particular piece of technology that is used in accordance with laptops and phones is headphones. From virtual meetings to online learning, headphones became essential. I have enjoyed using my headphones but I hate it when the battery levels deplete while using them. My JBL TUNE510BT headphones although have a slow rate of battery discharge against time which I would love to use mathematical modelling to understand its battery discharging dynamics.

Aim

The aim of this exploration is to use mathematical modelling to find a curve that fits the rate of my headphone battery discharge over time.

Rationale

I chose to explore this due to its applicability in real-life contexts and its direct relevance to my own experiences. As a frequent user of headphones, relying on them for various tasks throughout the day such as listening to music and receiving calls, the frustration of untimely battery depletion has often disrupted my workflow and leisure. Using mathematics, I would love to understand how my headphone battery discharges per unit time and even seek knowledge for not only myself but also headphone users to optimise usage patterns. Using mathematical principles such as modelling, I will reach my aim of **‘How to use mathematical modelling to find a curve that fits the rate of my headphone battery discharge over time’**

Exploration

Data collection

I collected my data by monitoring the battery performance of my JBL TUNE510BT headphones while systematically recording the battery level at regular intervals of one hour. Battery percentage was indicated on my phone which was wirelessly connected to the headphones.

Assumptions made while collecting data include:

- There is no external interference. The data assumes no external factors influenced the battery performance such as electromagnetic interference or software bugs.
- Uniform Battery Characteristics. Characteristics such as capacity and internal resistance are assumed to be constant over time to ensure consistency in performance measurements.

Table 1: JBL TUNE510BT Headphone Battery Performance Data (%) against Time (Hours)

Time (hours)	Battery level (%)
0	100
1	100
2	100
3	100
4	95
5	95
6	95
7	95
8	90
9	90
10	90
11	90
12	85
13	85
14	80
15	80
16	80
17	75
18	75
19	70
20	70
21	65
22	65
23	65
24	60
25	60
26	60
27	60
28	55
29	55
30	55

Data analysis

I can observe that the battery level percentage of my headphones is decreasing with an increase in time and considering that the pattern is not constant, it suggests that the relationship between these two variables is not directly proportional. I want to know which type of cause-and-effect relationship the variables exhibit be it inverse proportionality or even a more complex relationship through curve fitting. Involving mathematical models such as regression or exponential decay will enable me to analyse how one variable changes concerning the other with time.

The graphs were plotted and done using the Desmos Graphing calculator.

Linear function (Linear regression)

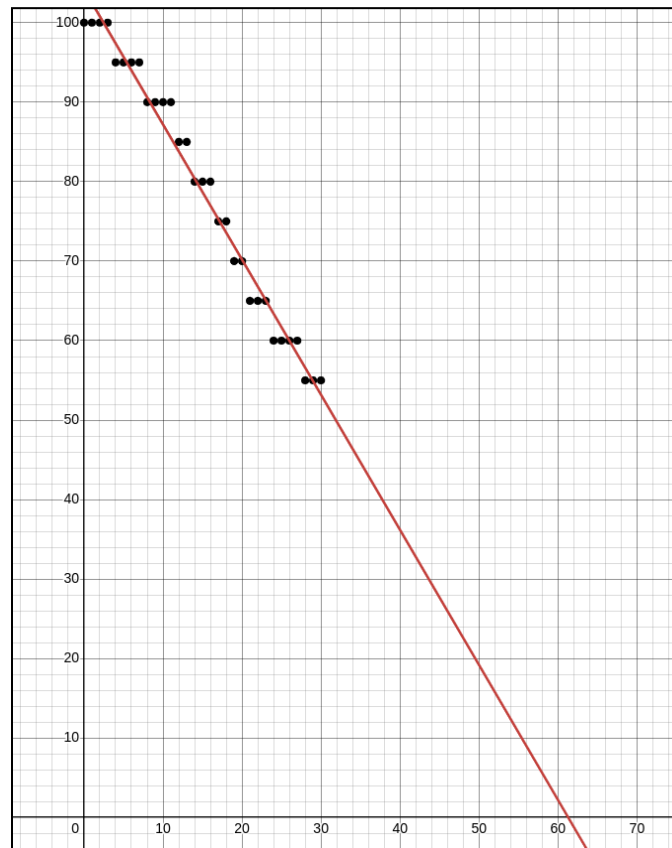


Figure 1: Graph of data points with a fitted linear function using Desmos

Firstly, the dataset demonstrates perfect collinearity between the independent variable, time and the dependent variable, battery level percentage. Time progresses uniformly from 0 to 30 hours

without variation, while the battery level remains constant for certain intervals. For instance, all battery level percentages endure for at least 2 hours and at most 4 hours such as 60%. As linear regression works on the assumption that the decrease is constant, it does not end up agreeing with my dataset as my Bluetooth headphone's battery level percentage is not decreasing uniformly (Duke University).

The linear regression equation derived from my dataset is $y = -1.6996x + 104.204$. It has a negative slope coefficient of (-1.700) which here represents the rate of change in the battery level per unit increases in time. However, the y-intercept is 104.204 which is problematic. In theory, it should represent the battery level of my JBL headphones when they are fully charged, i.e. 100%. Hence an intercept exceeding this threshold is unrealistic and undermines the real world practicality.

Therefore, my data does not support any relationship that can be determined through the use of linear regression as any model which should use linear regression should be bound with the following conditions:

$$y = mx + c \quad 0 < c \leq 100$$

[1]

Exponential functions



Figure 2: Graph of data points with a fitted exponential function using Desmos

The graph function is $f(x) = 106.336 \cdot 0.979^x$ represents the battery's charge level as a function of time x . In the function, the term 106.336 denotes the initial charge of the battery of the headphones when $x = 0$. This value acts as a vertical translation of the exponential function as it shifts the curve upwards or downwards from the x-axis. However, it is important to note that the battery charge levels are constrained between 0% and 100%. Therefore, having an initial charge exceeding 100% contradicts practical constraints and does not reflect how my Bluetooth headphones discharge with an increase in time. The base of the exponential function, 0.979, plays a crucial role in determining how rapidly the battery level diminishes over time.

Upon examining the model exponential graph, I observe a horizontal asymptote. The exponential curve gradually descends and seems to never intersect with the line $y = 0$, which represents the x-axis. This observation suggests that as time progresses infinitely, the battery's charge level approaches but never reaches zero. Conducting a limit analysis on line $y = 0$, could further elaborate the behaviour of the function as it approaches infinity allowing me to understand more about this.

The function does not meet the form $\frac{g(x)}{h(x)}$ I could not use L'Hopital's rule and had to find an alternative method to find the limit analysis of my curve.

The Equation of the horizontal asymptote of a function is given by;

$$y = \lim_{x \rightarrow \infty} f(x)$$

Through using the exponential rule that $x = e^{\ln(x)}$, I can take advantage of the fact that e_x and natural logarithm (ln) are inverse functions of each other and modify the exponential function ("10.3: Inverse Functions and Logarithms").

$$y = 106.336 (0.979)^x$$

$$y = 106.336 e^{\ln(0.979)^x}$$

$$y = 106.336 e^{\ln(0.979) \cdot x}$$

[2]

I can solve $\ln(0.979)$ separately which I get:

$$\begin{aligned} \ln(0.979) &= -0.02122364 \\ &\approx -0.021 \quad (3 \text{ decimal places}) \end{aligned}$$

Making the function result in:

$$y = 106.336 e^{(-0.02) \cdot x}$$

Therefore the horizontal asymptote of my function will be:

$$y = \lim_{x \rightarrow \infty} 106.336 e^{-(0.02x)}$$

$$y = 106.336 e^{-(0.02) \times \infty}$$

[3]

When e is raised to power negative infinity, the resulting value decreases significantly and becomes exceedingly small thus it tends towards zero ie. $e^{-\infty} = 0$. This makes the function $y = 0$.

This indicates that the line $y = 0$ is the horizontal asymptote of my function which indicates that as the battery discharge function progresses towards infinite time, it tends towards zero. Although using real-life practicality, time should tend toward a finite time as the headphones will eventually discharge fully at a particular point making the exponential function a bit questionable as it cannot answer that.

Interpolation

Gregory Newton Forward Difference interpolation

Exponential and linear functions have failed to model how my JBL TUNE500BT headphones discharge with time. I now want to generate a polynomial function that can validate my dataset and observe the analytical characteristics of the graph. To meet my aim I sought out alternative methods which include interpolation. Through interpolation, using a set of numerical data of coordinates with equal or unequal class sizes, one can find the functional value of an intermediate x value (The Editors of Encyclopaedia Britannica).

My data set has equal intervals in the independent variable, time, hence I need an interpolation method that fits my data. Since my independent variable is equidistant this rules out Lagrange's interpolation as it is for unequal class size data (Talukdar). Therefore, the best interpolation method for me to use is Gregory Newton's Difference interpolation. There are forward and backward differences, since my chosen data point is 7.5 hours and is closer to 0 than 30 hours, the easier method to use is Gregory Newton's Forward Difference interpolation.

31 sets of data is a lot and would be prone to errors therefore I used my data set in intervals of 5 to find the corresponding value using the interpolation method.

Table 2: Battery Performance Data for my JBL TUNE510BT Headphones (% Charge Remaining) Over Time in 5-Hour Intervals

Time (Hours)	0	5	10	15	20	25	30
Battery level (%)	100	95	90	80	70	60	55

Table 3: Forward Difference Table for my JBL TUNE510BT Headphones' Battery Performance (% Charge Remaining) Over Time (5-Hour Intervals)

x - Time	y - Battery Level	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
0	100						
		-5					
5	95		0				
		-5		-5			
10	90		-5		10		
		-10		5		-15	
15	80		0		-5		25
		-10		0		10	
20	70		0		5		
		-10		5			
25	60		5				
		-10					
30	55						

I used the formula:

$$f(x) = Y_0 + P \cdot \Delta y + \frac{P(P-1)}{2!} \cdot \Delta^2 Y + \frac{P(P-1)(P-2)}{3!} \cdot \Delta^3 Y + \dots$$

[4]

Where:

$$P = \frac{X - X_0}{h}$$

X = x-value of requirement (7.5)

h = interval size

X₀ = initial x-value

Y₀ = initial y-value

$$P = \frac{X - 0}{5} = \frac{X}{5}$$

$$\begin{aligned} f(x) = & 100 + \frac{X}{5}(-5) + \frac{\frac{X}{5}(\frac{X}{5}-1)}{2!} \cdot (0) + \frac{\frac{X}{5}(\frac{X}{5}-1)(\frac{X}{5}-2)}{3!} \cdot (-5) + \frac{\frac{X}{5}(\frac{X}{5}-1)(\frac{X}{5}-2)(\frac{X}{5}-3)}{4!} \\ & \cdot (10) + \frac{\frac{X}{5}(\frac{X}{5}-1)(\frac{X}{5}-2)(\frac{X}{5}-3)(\frac{X}{5}-4)}{5!} \cdot (-15) + \frac{\frac{X}{5}(\frac{X}{5}-1)(\frac{X}{5}-2)(\frac{X}{5}-3)(\frac{X}{5}-4)(\frac{X}{5}-5)}{6!} \cdot (-25) \end{aligned}$$

[5]

$$\begin{aligned} f(x) = & 100 - x - \frac{1}{150}x^3 + \frac{1}{10}x^2 - \frac{1}{3}x + \frac{1}{1500}x^4 - \frac{1}{50}x^3 + \frac{11}{60}x^2 - \frac{1}{2}x - \frac{1}{25000}x^5 + \\ & \frac{1}{500}x^4 + \frac{7}{200}x^3 + \frac{1}{4}x^2 - \frac{3}{5}x + \frac{1}{450000}x^6 - \frac{1}{6000}x^5 + \frac{17}{3600}x^4 - \frac{1}{16}x^3 + \frac{137}{360}x^2 - \\ & \frac{5}{6}x \end{aligned}$$

$$f(x) = 100 - \frac{49}{15}x - \frac{149}{1200}x^3 + \frac{329}{360}x^2 + \frac{133}{18000}x^4 - \frac{31}{150000}x^5 + \frac{1}{450000}x^6$$

[6]

Using Equation 6, I generated a fitted curve. Upon plotting the polynomial, the resulting curve closely matched my data points indicating a significant alignment between the model and dataset with an x boundary from 0 to 30 i.e. $\{0 \leq x \leq 30\}$.

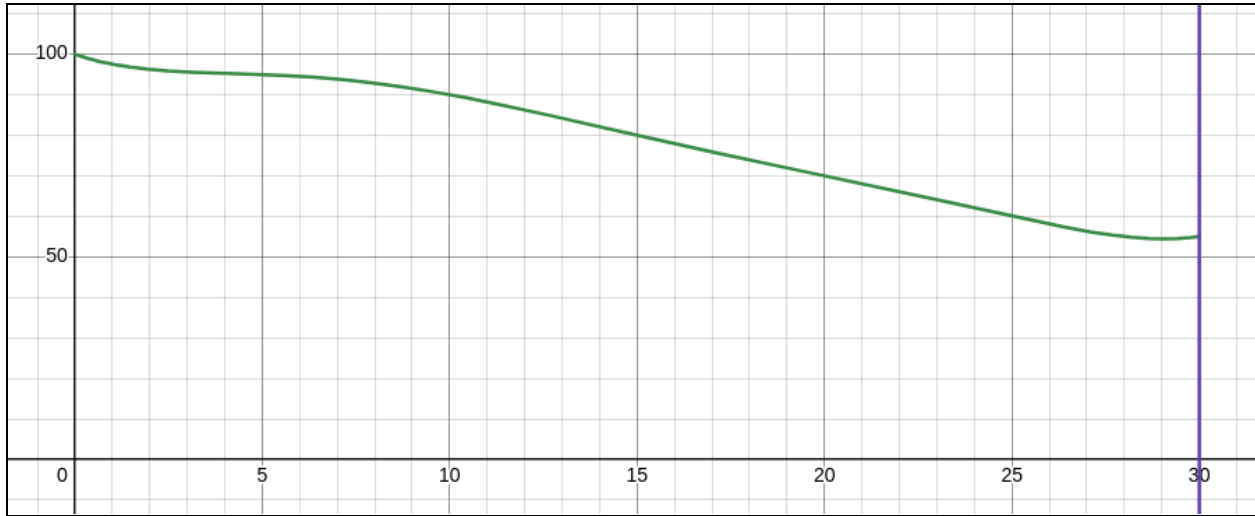


Figure 3: Data Points with Fitted Curve Using Newton's Forward Interpolation of 6th Order Exponential Function (Generated with Desmos)

$$f(x) = 100 + (-7.5) + (0.3125) + (0.234375) + (0.17578125) + (0.1708984375)$$

$$f(x) = 93.3935469$$

$$f(x) \approx 93.394 \text{ (3 decimal places)}$$

The graph intersects a pivotal point (100,0) which aligns with the full charge state of my headphones at time $t = 0$. This alignment with real-life logic proves that I can now dismiss both the exponential and linear functions. However, I acknowledge a potential source of error due to my usage of 5-hour intervals for time instead of the more precise 1-hour interval. This introduces translation errors as I used wider intervals of time that may result in missing variations or fluctuations in the data. The creation of a 30th order polynomial looked very tedious but I started to work on it to meet my initial aim '***to use mathematical modelling to find a curve that fits the rate of my headphone battery discharge over time***'. I tried it on paper but it became too much and I therefore devised a javascript code that could accept my dataset and tell me the intermediate value of y when x is 7.5 (see Appendix 1).

Despite the code failing numerous times, I have been able to solve it and found data that showed using the 5-hour interval was prone to errors as I previously concluded. Using the code I found

the corresponding value of y when $x = 7.5$ is approximately 87.715 (**3 decimal places**). The difference between using a 6th-order polynomial and a 30th-order polynomial is:

$$\text{Error} = y \text{ value of } 6^{\text{th}} \text{ polynomial} - y \text{ value of } 30^{\text{th}} \text{ polynomial}$$

$$\text{Error} = 93.394 - 87.715$$

$$\text{Error} = 5.679$$

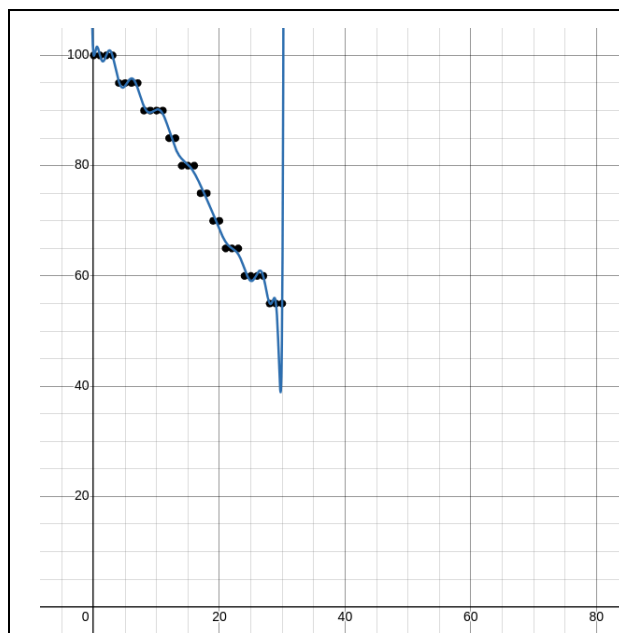
To calculate the percentage error, I used the formula:

$$\text{Percentage Error} = \frac{\text{Error}}{\text{Actual Value}} \times 100\%$$

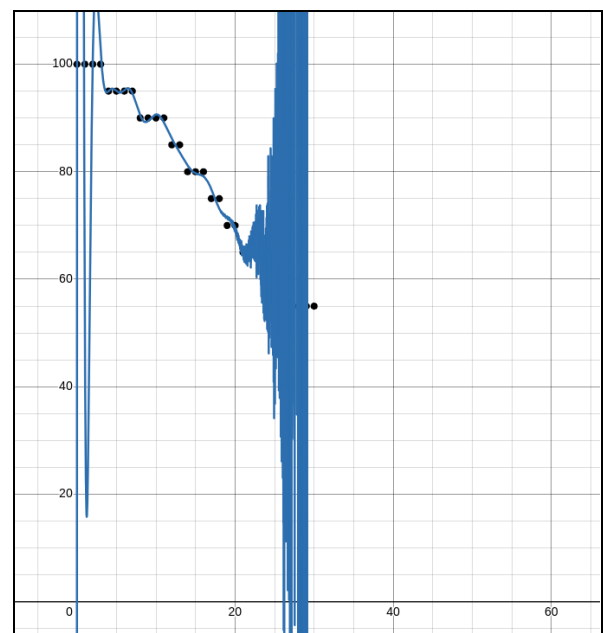
$$\text{Percentage Error} = \frac{5.679}{87.715} \times 100\% \approx 6.474\%$$

The error calculation does confirm that the 30th polynomial provides a more accurate approximation of the function at that specific point than the 6th polynomial order. Upon manipulating the code (see Appendix 1), I was able to express the entire polynomial.

I then used the regression function for the actual data point and tried fitting it into a polynomial of the 30th order to get more price results and I was able to obtain the following graphs;



*Figure 4: Data Points with Fitted Curve
Using Newton's Forward Interpolation of
21st Order Exponential Function (Generated
with Desmos)*



*Figure 5: Data Points with Fitted Curve
Using Newton's Forward Interpolation of
26th Order Exponential Function (Generated
with Desmos)*

Upon thorough analysis of my graphs, I initially had doubts regarding their accuracy. High-degree interpolation polynomials with uniformly distributed points have oscillatory behavior around the edges which can be observed in Figure 5. This is known as the Runge phenomenon, a consequence of employing Newton Interpolation techniques on many equidistant data points thus resulting in oscillations at the edges of the dataset (Epperson). Newton Forward Interpolation could not help me meet my aim and therefore I ended up using piecewise functions to break down my data points into smaller subsets to meet my aim.

Piecewise interpolation

Data was then divided into subparts of 5 x and y pairs apart from one which would have 6 as the total number of data points is 31.

First 5 data subset

(0,100), (1,100), (2,100), (3,100), (4,95),

x - Time	y - Battery Level	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	100				
		0			
1	100		0		
		0		0	
2	100		0		-5
		0		-5	
3	100		-5		
		-5			
4	95				

I can observe that $P=X$ as;

$$P = \frac{X-X_0}{h}$$

In my data set in Table 1, $X_0 = 0$ and $h = 1$, therefore;

$$P = \frac{X-0}{1} = \frac{X}{1} - \frac{0}{1} = X$$

$$\therefore P = X$$

Equation 4 follows the binomial theorem pattern and therefore here I can deduce the Equation to be:

$$y(x) = \sum_{n=0}^{m-1} \frac{x!}{(x-n)!n!} \Delta^n y$$

[7]

$$y(x) = \sum_{n=0}^{m-1} (xCn) \Delta^n y$$

[8]

Where;

m= total number of data points

x = desired value of interpolation

$\Delta^n y$ = nth forward difference coefficient

$$P_1(x) = \sum_{n=0}^4 (xCn) \Delta^n y$$

$$P_1(x) = (xC0)y_0 + (xC1)\Delta y + (xC2)\Delta^2 y + (xC3)\Delta^3 y + (xC3)\Delta^4 y$$

$$P_1(x) = (xC0) \cdot 100 + (xC1) \cdot 0 + (xC2) \cdot 0 + (xC3) \cdot 0 + (xC3) \cdot -5$$

$$P_1(x) = 100 - \frac{5}{24}x^4 + \frac{5}{4}x^3 - \frac{55}{24}x^2 + \frac{5}{4}x$$

[9]

Second 5 data subset

(5,95), (6,95), (7,95), (8,90), (9,90)

x - Time	y - Battery Level	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
5	95				
		0			
6	95		0		
		0		-5	
7	95		-5		15
		-5		10	
8	90		5		
		0			
9	90				

$$P_2(x) = \sum_{n=5}^9 (xCn - 5) \Delta^{n-5} y$$

$$P_2(x) = (xC0)y_o + (xC1)\Delta y + (xC2)\Delta^2 y + (xC3)\Delta^3 y + (xC3)\Delta^4$$

$$P_2(x) = (xC0) \cdot 95 + (xC1) \cdot 0 + (xC2) \cdot 0 + (xC3) \cdot -5 + (xC3) \cdot 15$$

$$P_2(x) = 95 + \frac{x(x-1)(x-2)}{3!} \cdot (-5) + \frac{x(x-1)(x-2)(x-3)}{4!} \cdot (15)$$

$$P_2(x) = \frac{5}{8}x^4 - \frac{55}{12}x^3 + \frac{75}{8}x^2 - \frac{65}{12}x + 95$$

[10]

The required polynomial is $P_2(x - 5)$ therefore;

$$P_2(x - 5) = \frac{5}{8}(x - 5)^4 - \frac{55}{12}(x - 5)^3 + \frac{75}{8}(x - 5)^2 - \frac{65}{12}(x - 5) + 95$$

$$P_2(x) = \frac{5}{8}x^4 - \frac{205}{12}x^3 + \frac{1375}{8}x^2 - \frac{9065}{12}x + 1320$$

[11]

Third 5 data subset

(10,90), (11,90), (12,85), (13,85), (14,80)

x - Time	y - Battery Level	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
10	90				
		0			
11	90		-5		
		-5		10	
12	85		5		-20
		0		-10	
13	85		-5		
		-5			
14	80				

$$P_3(x) = \sum_{n=10}^{14} (xCn - 10) \Delta^{n-10} y$$

$$P_3(x) = (xC0)y_0 + (xC1)\Delta y + (xC2)\Delta^2 y + (xC3)\Delta^3 y + (xC3)\Delta^4$$

$$P_3(x) = (xC0) \cdot 90 + (xC1) \cdot 0 + (xC2) \cdot -5 + (xC3) \cdot 10 + (xC3) \cdot -20$$

$$P_3(x) = 90 + \frac{x(x-1)}{2!} \cdot (-5) + \frac{x(x-1)(x-2)}{3!} \cdot 10 + \frac{x(x-1)(x-2)(x-3)}{4!} \cdot (-20)$$

$$P_3(x) = -\frac{5}{6}x^4 = \frac{20}{3}x^3 - \frac{50}{3}x^2 + \frac{65}{6}x + 90$$

[12]

The required polynomial is $P_2(x - 10)$ therefore;

$$P_3(x - 10) = -\frac{5}{6}(x - 10)^4 + \frac{20}{3}(x - 10)^3 - \frac{50}{3}(x - 10)^2 + \frac{65}{6}(x - 10) + 90$$

$$P_2(x) = -\frac{5}{6}x^4 + 40x^3 - \frac{2150}{3}x^2 + \frac{11355}{2}x - 16685$$

[13]

Fourth 5 data subset

(15,80), (16,80), (17,75), (18,75), (19,70)

x - Time	y - Battery Level	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
15	80				
		0			
16	80		-5		
		-5		10	
17	75		5		-20
		0		-10	
18	75		-5		
		-5			
19	70				

$$P_4(x) = \sum_{n=15}^{19} (xCn - 15) \Delta^{n-15} y$$

$$P_4(x) = (xC0)y_0 + (xC1)\Delta y + (xC2)\Delta^2 y + (xC3)\Delta^3 y + (xC3)\Delta^4$$

$$P_4(x) = (xC0) \cdot 80 + (xC1) \cdot 0 + (xC2) \cdot -5 + (xC3) \cdot 10 + (xC3) \cdot -20$$

$$P_4(x) = 80 + \frac{x(x-1)}{2!} \cdot (-5) + \frac{x(x-1)(x-2)}{3!} \cdot 10 + \frac{x(x-1)(x-2)(x-3)}{4!} \cdot (-20)$$

$$P_4(x) = -\frac{5}{6}x^4 + \frac{20}{3}x^3 - \frac{50}{3}x^2 + \frac{65}{6}x + 80$$

[14]

The required polynomial is $P_2(x - 15)$ therefore;

$$P_3(x - 15) = -\frac{5}{6}(x - 10)^4 + \frac{20}{3}(x - 10)^3 - \frac{50}{3}(x - 10)^2 + \frac{65}{6}(x - 10) + 80$$

[15]

Fifth 5 data subset

x - Time	y - Battery Level	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
20	80				
		-5			
21	80		5		
		0		-5	
22	75		5		0
		0		-5	
23	75		-5		
		-5			
24	70				

$$P_5(x) = \sum_{n=20}^{24} (xCn - 20) \Delta^{n-20} y$$

$$P_5(x) = (xC0)y_0 + (xC1)\Delta y + (xC2)\Delta^2 y + (xC3)\Delta^3 y + (xC3)\Delta^4$$

$$P_5(x) = (xC0) \cdot 70 + (xC1) \cdot -5 + (xC2) \cdot 5 + (xC3) \cdot -5 + (xC3) \cdot 0$$

$$P_5(x) = 70 + x \cdot (-5) + \frac{x(x-1)}{2!} \cdot (5) + \frac{x(x-1)(x-2)}{3!} \cdot 5$$

$$P_5(x) = -\frac{5}{6}x^3 + 5x^2 - \frac{55}{6}x + 70$$

[15]

The required polynomial is $P_5(x - 20)$ therefore;

$$P_5(x - 15) = -\frac{5}{6}(x - 20)^3 + 5(x - 20)^2 - \frac{55}{6}(x - 20) + 70$$

[16]

Sixth 6 data subset

x - Time	y - Battery Level	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
25	60					
		0				
26	60		0			
		0		-5		
27	60		-5		15	
		-5		10		-30
28	55		5		-15	
		0		-5		
29	55		0			
		0				
30	55					

$$P_6(x) = \sum_{n=25}^{30} (xCn - 25) \Delta^{n-25} y$$

$$P_6(x) = (xC0)y_0 + (xC1)\Delta y + (xC2)\Delta^2 y + (xC3)\Delta^3 y + (xC4)\Delta^4 + (xC5)\Delta^5$$

$$P_6(x) = (xC0) \cdot 60 + (xC3) \cdot -5 + (xC4) \cdot 15 + (xC4) \cdot -30$$

$$P_6(x) = 60 + \frac{x(x-1)(x-2)}{3!} \cdot -5 + \frac{x(x-1)(x-2)(x-3)}{4!} \cdot (15) + \frac{x(x-1)(x-2)(x-3)(x-4)}{5!} \cdot (-30)$$

$$P_6(x) = 60 - \frac{40}{3}x^3 + \frac{175}{8}x^2 - \frac{137}{12}x + \frac{25}{8}x^4 - \frac{1}{4}x^5$$

[17]

The required polynomial is $P_2(x - 25)$ therefore;

$$P_6(x - 25) = 60 - \frac{40}{3}(x - 25)^3 + \frac{175}{8}(x - 25)^2 - \frac{137}{12}(x - 25) + \frac{25}{8}(x - 25)^4 - \frac{1}{4}(x - 25)^5$$

[18]

After doing each piecewise function separately I noted them down to put them in the Desmos graphing calculator to get a continuous function.

$$P(x) = \begin{cases} \frac{5}{24}x^4 + \frac{5}{4}x^3 - \frac{55}{24}x^2 + \frac{5}{4}x + 100 & \forall x \in [0, 1, 2, 3, 4] \\ \frac{5}{8}x^4 - \frac{205}{12}x^3 + \frac{1375}{8}x^2 - \frac{9065}{12}x + 1320 & \forall x \in [5, 6, 7, 8, 9] \\ -\frac{5}{6}x^4 + 40x^3 - \frac{2150}{3}x^2 + \frac{11355}{2}x - 16685 & \forall x \in [10, 11, 12, 13, 14] \\ -\frac{5}{6}x^4 + \frac{170}{3}x^3 - \frac{4325}{3}x^2 + \frac{97565}{6}x - 68520 & \forall x \in [15, 16, 17, 18, 19] \\ -\frac{5}{6}x^3 + 56x^2 - \frac{7255}{6}x + 8920 & \forall x \in [20, 21, 22, 23, 24] \\ -\frac{1}{4}x^5 + \frac{275}{8}x^4 - \frac{5665}{3}x^3 + \frac{414425}{8}x^2 - \frac{8516387}{12}x + 3884460 & \forall x \in [25, 26, 27, 28, 29, 30] \end{cases}$$

Conclusion

This exploration has taken me on a journey through various mathematical concepts, allowing me to connect classroom learning with real-world applications. I have been able to see how the binomial theorem, sigma summation and factorial in interpolation methods while looking into new mathematical ideas along the way. Through this, I have expanded my understanding of math and gained insight into how modelling can be done. In the end, I achieved the aim of my exploration which is to **'use mathematical modelling to find a curve that fits the rate of my headphone battery discharge over time'** which can be observed in Figure 7.

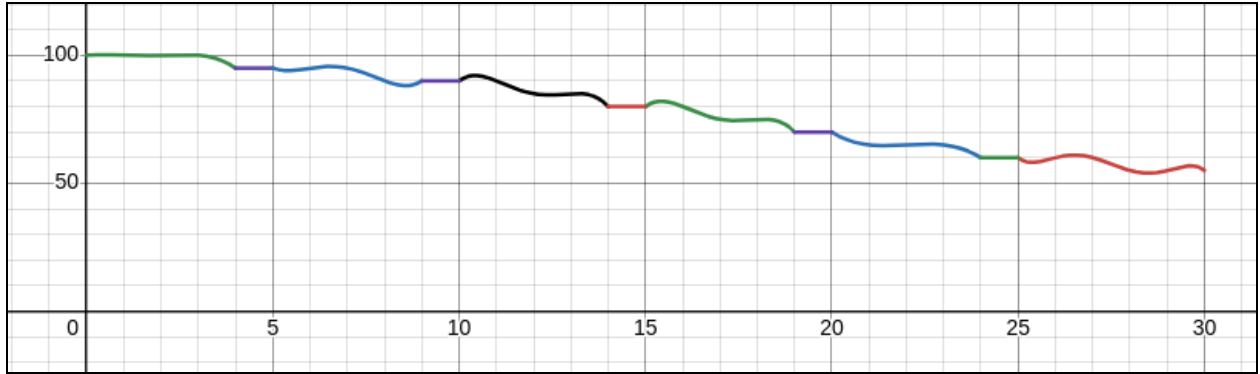


Figure 6: Graph illustrating the piecewise function representing the discharge of my headphone battery over time

Evaluation

The strength of using piecewise interpolation is that it eliminates the Runge phenomenon making data more accurate among terminal points of my data set making it more reliable on the end and start point the Newton's forward method of interpolation. Limitations of the exploration included that the use of a piece-wise function limits the use of one compacted single fraction to represent all data points. Also, the interpolation of huge data sets brings about distortions and anomalies that make it nearly impossible to find an accurate interpolating polynomial for the data.

An extension of my exploration could include the use of cubic spline interpolation to determine the best curve fit for the polynomial function and the use of Fourier decomposition to add insights into the underlying patterns. Cubic spline interpolation would provide smooth curves that pass through all points addressing noisy data while the Fourier decomposition allows the analysis of periodic patterns within the dataset that may reveal hidden periodic or oscillatory trends.

Appendix

Appendix 1: Implementation Code for Newton's Forward Difference Method

//x => x values y=> y values n=> no. of data points i => index of number j=> order of forward difference

```
function forwardDifference(x, y, n, i, j) {  
  if (j === 0) {  
    return y[i];  
  } else {  
    return forwardDifference(x, y, n, i + 1, j - 1) - forwardDifference(x, y, n, i, j - 1);  
  }  
}
```

//value =>interpolated value to be used

//p => (X-Xnote)/h

//the loop runs from j when it is 0 to n-1

//k represents the index of the current term within the divided difference expression for the order j.

//The Horner's method is used for precision

```
function coefficients(x, y, n, value) {  
  const h = x[1] - x[0];  
  const p = (value - x[0]) / h;  
  let result = forwardDifference(x, y, n, 0, 0);  
  
  for (let j = 1; j < n; j++) {  
    let term = forwardDifference(x, y, n, 0, j);  
    let coeff = 1;  
  
    for (let k = 0; k < j; k++) {  
      coeff *= (p - k) / (k + 1);  
    }  
    result += term * coeff;  
  }  
  return result;  
}
```

//Node js readline module to make the CLI interactive

```
function interface() {  
  const readline = require('readline');  
  const rl = readline.createInterface({  
    input: process.stdin,
```

```

    output: process.stdout
  });

  rl.question("Enter the number of data points: ", function(n) {
    let x = [];
    let y = [];
    const readX = (i) => {
      if (i < n) {
        rl.question(`Enter the value of x${i}: `, function(value) {
          x.push(parseFloat(value));
          readY(i);
        });
      } else {
        readValue();
      }
    };
    const readY = (i) => {
      rl.question(`Enter the value of y${i}: `, function(value) {
        y.push(parseFloat(value));
        readX(i + 1);
      });
    };
    const readValue = () => {
      rl.question("Enter the value for interpolation: ", function(value) {
        let result = coefficients(x, y, n, parseFloat(value));
        console.log(`Interpolated value at ${value} is ${result.toFixed(3)}`);
        rl.close();
      });
    };
    readX(0);
  });
}

interface();

```

Works Cited

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